1 Socio-Pragmatic Architecture and System Component Specifications

1.1 Pragmatics, Level-k Reasoning, and Searlean Realities

To model social networks where agents think about other agents' thoughts, a custom Probabilistic Domain Specific Language (DSL) was developed. This language parses recursively nested mental states (`thinks`, `chooses`, `knows`, `observes`) to simulate Rational Speech Act (RSA) matrices and level-k strategic convergence (e.g., Schelling games, political and religious belief polarization networks).

To satisfy John Searle's philosophy of the social world, the language was expanded to Grammar 3, embedding a dedicated, constitutive status operator:

$$\mathcal{D}_{\text{inception}}\Big(\langle C,\, X,\, Y\rangle\Big) \implies (X \text{ counts as } Y \text{ in } C) \tag{1}$$

This allows the engine to instantiate institutional facts that cannot be measured via physical sensory inputs. The grammar cleanly separates illocutionary forces into five distinct categories: Assertives, Directives, Commissives, Expressives, and Declarations. This framework manages security perimeters, access control, and legal obligations containing strict temporal deadlines (t_{deadline}).

1.2 Specification and System Components

The neuro-symbolic system acts as a bounded, type-safe transducer. It inputs structured strings or queries and produces learned contexts, classifications, or deductive responses based on the available training material.

The system is powered by seven interlocking components:

1. **LLM as Co-Designer:** Acts as an automated reward stencil synthesizer and meta-level instructor.
2. **Python Orchestrator:** The master continuous execution plane and virtual machine runner.
3. **Forest of Decision Trees:** The core associative, hierarchical memory array for learning situational contexts.
4. **The Prolog Guardrail (`kb.pl`):** The formal logical attention filter ensuring type-safe branch splits.
5. **RL with DSLs:** Sequential decision-making engines optimizing policies over bounded continuous languages.
6. **Automated Algebraic Solver:** Symmetric Group (S_3) controllers providing invertible, loss-free data rollbacks.
7. **Graph Neural Network (GNN):** PyTorch Geometric modules processing large-scale empirical church network graphs.

2 Context Layer: Interlocking Decision Forests

2.1 Decision Trees as Associative Agent Memory

Decision trees function as the agent's structural memory matrix and are used directly for context classification. The forest learns complex relational concepts as presented by sparse multi-modal exemplars. The non-terminal internal nodes are derived dynamically from the exemplars, evaluating testing features or structured parthood (`part_of`) relationships.

The decision tree recursion is explicitly configured to support nested conceptual compositionality, enabling the system to learn high-level meta-concepts strictly in terms of simpler, underlying local sub-concepts. Concepts are formulated in LISP S-expressions and can be indefinitely nested. Terminal leaf nodes retain partial exemplar arrays and domain-specific solution techniques. There are multiple independent trees trained for each category of concepts, ensuring that computationally analogous scenarios converge cleanly onto the exact same leaf node.

2.2 `kb.pl` Logic as the Attention Guardrail

The Prolog logic layer operates as an explainable attention filter. If a decision tree attempts to shift its focus or generate a new, random feature split to minimize statistical error over noisy data, the `kb.pl` file audits that choice. It blocks the tree from chasing noise or hallucinating branches, verifying that variables are legally and structurally meaningful to the macro-institution.

3 Training and Inferencing Plane

3.1 Reduced-Vocabulary Intrusion Detection Gate

To protect the system from un-vetted adversarial payloads, the LLM provides all prompts in a strictly reduced vocabulary form to the neuro-symbolic intrusion detection system (IDS). This system was engineered in a previous development phase; it is maintained as a decoupled, upstream filter boundary to allow current research to focus exclusively on multi-agent system (MAS) consensus networks and parallel branching poset architectures without character cross-talk.

3.2 Inductive Ontology Replacement and Naming Realignment

Our architecture replaces manual ontology revision entirely by utilizing the unique mathematical properties of learned decision trees, rendering the shared strategy completely *inductively definable*. The 5-step merging procedure unrolls at the target agent as follows: (1) Logic Basis Consensus on the received `kb.pl` file; (2) Model Payload Buffering of the source tree; (3) Logical Interrogation and Repair by stripping type names to blind queries; (4) Inductive Retraining Pass over updated exemplars; and (5) Ontology Substitution for a conflict-free semantic merger.

3.3 Symmetric Cyber-Physical Inductive Modeling and High-Fidelity EMT Grounding

We formalize a completely symmetric framework for cyber-physical inductive modeling, ensuring identical mathematical authority whether deployed from the AI data center perspective or within the power utility's SCADA dispatch perimeter. To ground macroscopic synchronization behaviors into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical

simulations are performed over a modified IEEE 39-bus test system using the open-source Julia platform `PowerSimulationsDynamics.jl`. Let the transient state of an electrical node i be characterized by a continuous differential-algebraic power vector $\boldsymbol{\xi}_i(t)$:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

where $V_i(t)$ represents the instantaneous voltage magnitude, $\delta_i(t)$ defines the phase angle deviation, and $P_{\text{IBR},i}(t), Q_{\text{IBR},i}(t)$ monitor the active and reactive power outputs of local inverter-based renewable resources.

These continuous wavelets are processed natively by our decentralized inductive learners as extensional grounding exemplars (`exs.pl`) sampled at a resolution of $\Delta t = 0.5$ ms. This allows the framework to detect microscopic transient voltage droops ($V_i(t) < 0.95$ p.u.) and enforce strict Sherman-Gable (1983) CSMA/CD delay gates before sub-synchronous oscillations propagate across the complete forcing amplitude spectrum:

3.3.1 Phase 1: Open-Loop Repetitive Forcing (Unstable Chaos Cloud)

```
FORCING AMP | ASYMPTOTIC FREQUENCY PEAKS (Hz)  | REGIME
-------------------------------------------------------------------
 0.00 GW    | 24.965, 24.966, 24.980...       | [CHAOTIC ATTRACTOR]
 2.14 GW    | 21.438, 21.509, 21.916...       | [CHAOTIC ATTRACTOR]
 4.29 GW    | 18.844, 18.871, 18.885...       | [CHAOTIC ATTRACTOR]
 6.43 GW    | 15.720, 15.776, 16.024...       | [CHAOTIC ATTRACTOR]
 8.57 GW    | 12.747, 12.878, 12.886...       | [CHAOTIC ATTRACTOR]
10.71 GW    | 9.715, 9.774, 9.893...          | [CHAOTIC ATTRACTOR]
12.86 GW    | 6.677, 6.684, 6.714...          | [CHAOTIC ATTRACTOR]
15.00 GW    | 3.681, 3.698, 3.714...          | [CHAOTIC ATTRACTOR]
-------------------------------------------------------------------
```

3.3.2 Phase 2: Closed-Loop MARL Gating (Phase-Locked Steady State)

```
FORCING AMP | ASYMPTOTIC FREQUENCY PEAKS (Hz)  | REGIME
-------------------------------------------------------------------
 0.00 GW    | 24.965                          | [PHASE-LOCKED STATE]
 2.14 GW    | 24.964                          | [PHASE-LOCKED STATE]
 4.29 GW    | 24.971                          | [PHASE-LOCKED STATE]
 6.43 GW    | 24.954                          | [PHASE-LOCKED STATE]
 8.57 GW    | 24.965                          | [PHASE-LOCKED STATE]
10.71 GW    | 24.942                          | [PHASE-LOCKED STATE]
12.86 GW    | 24.948                          | [PHASE-LOCKED STATE]
15.00 GW    | 24.943                          | [PHASE-LOCKED STATE]
-------------------------------------------------------------------
```

3.4 The Temporal Sequence of Model Collision and Dialectical Self-Repair

We resolve the temporal boundaries governing theory deconfliction by establishing that *model collision occurs strictly during the process of attempting to reach consensus*, serving as the direct operational trigger for dialectical self-repair. If an incoming abstract tree payload from a source agent maps to a premise configuration that contradicts the target node's immutable invariants, a model collision is instantiated at the boundary gateway.

Rather than executing a local overwrite or allowing system trivialization, the target agent implements Philippe Besnard and Anthony Hunter's deductive argumentation framework to automatically construct an active *Undercutter* counter-argument tuple:

$$\mathcal{A}_{\text{counter}} = \langle \Phi_{\text{target}}, \neg\phi_{\text{source}} \rangle \tag{3}$$

This undercutter payload is transmitted back across the Model Context Protocol (MCP) gateway to the origin source node. The source agent treats its previous hypothesis as defeated, triggering immediate local back-propagation pruning across its decision forest to eliminate the anomalous feature splits. Once the source node completes this structural repair, the target node executes its localized inductive retraining pass over the aligned exemplar matrix, culminating in a sound, unified strategic consensus across the decentralized network topology.

3.5 Network Bandwidth Optimization via Abstract Model Compression Invariants

Fulfilling multi-agent coordinate alignment across a distributed network of $10,000$ active nodes presents an asymptotic communication bottleneck under raw data transport regimes. Transmitting raw, high-dimensional extensional exemplar matrices generates prohibitive network overhead that saturates communication buses and introduces data privacy vulnerabilities. Following the local network data routing paradigms of **Sherman & Gable (1983)**, our framework guarantees optimal channel efficiency by transmitting the compiled decision forest model as a compressed symbolic S-expression payload.

The local inductive tree generation pass operates as a loss-free relational filter that eliminates irrelevant exemplars, measurement noise, and non-bifurcating operational parameters. The network transmission bandwidth cost $\mathcal{C}$ scales cleanly as a function of graph depth d rather than data volume size:

$$\mathcal{C}(\mathcal{T}) \propto \mathcal{O}(d \cdot \Sigma_{\text{tokens}}) \ll \mathcal{O}(|\mathcal{E}| \cdot \text{Dim}_{\text{features}}) \tag{4}$$

By moving the compact structural compression model instead of raw grounding matrices, the multi-agent perimeter executes real-time blind interrogation and nomenclature deconfliction while conserving network traffic capacity and maintaining absolute data firewall boundaries.

3.6 Black-Hunter-Pan Dialogue Game Strategies Over Ontological Lattices

To structure the actual communication protocol between sovereign entities without database exposure, agents execute a preference-ordered dialogue game based on the algebraic principles of **Black, Hunter & Pan**. Let $\mathcal{D}_t$ represent the turn-taking state history of the dialogue at step t. The move strategy selection function $\mathcal{S}_x(\mathcal{D}_t)$ acts as a deterministic, preference-ordered selection transducer over the Description Logic poset, enforcing a strict hierarchical preference ranking:

$$\mathcal{S}_x(\mathcal{D}_t) = \text{Scounter}_x(\mathcal{D}_t) \succ \text{Sarg}_x(\mathcal{D}_t) \succ \text{Sconcede}_x(\mathcal{D}_t) \succ \langle x, \text{close}, \text{Topic}(\mathcal{D}_t) \rangle \tag{5}$$

where Scounter_x computes novel undercutter updates, Sarg_x handles active topic assertions, and Sconcede_x provides relevant semantic concessions derived from localized ontologies Δ_x. This configuration allows nodes to resolve structural conflicts, identify nomenclature variances, and establish an undefeated consensus path under full privacy preservation.

3.7 Edge-Grid Prosumer LAN Architectures and Consumer IoT Integration

The decentralized topology scales downward with structural parity to govern the explosive growth of consumer internet of things (IoT) edge devices, utility aggregators, and residential vehicle-to-home (V2H) bidirectional backup storage. Under conventional control regimes, external virtual power plants (VPPs) require un-vetted, direct centralized actuation over domestic parameters (e.g., raw thermostat manipulation or uncoordinated battery cell deep-discharging). This architecture bypasses these direct intervention vulnerabilities by treating the individual consumer property—containing Tesla rooftop solar panels, local battery storage bank primitives, and bidirectional automotive buffer systems—as a self-contained, firewall-protected Local Area Network (LAN) edge node.

The home network connects to the utility distribution bus exclusively through a single, secure gateway node executing the symmetric throughput invariant (S) under a responsive propagation parameter $a = 0.01$. Internal control variables, device lifetime margins, and private load habits are completely abstracted; the external interchange provider merely submits demand requests over thin MCP channels. Local inductive decision trees evaluate the residential state, and a Black-Hunter-Pan dialogue poset deconfliction pass negotiates the absolute power boundary exchange, protecting domestic structural sovereignty while providing certified stabilization to the inter-area grid.

3.8 Hierarchical Pattern Recognition of Complex Overhauled Mereological Objects

To fulfill automated active system identification over a complete regional topology without monolithic data dependencies, the learning forests parse the macro-grid as a complex mereological object. Rather than processing the network as an unstructured vector array, the system models the environment as an interlocking assembly of constituent sub-device classes bound by formal structural parthood (`part_of`) predicates.

The time-series input-output frames harvested from the modified IEEE 39-bus topology are segmented into fixed-length windows and passed to a scalable bi-level optimization core coupling the Prediction Error Method (PEM) with a Moving Horizon Estimation (MHE) state estimator as pioneered by **Zhou & Köhler (2024)**. By eliminating parameter-wide sensitivity matrices via an adjoint reformulation, the framework isolates physical constants and neural network residuals with up to a 215× speedup.

The recovered parameters across the 5 Synchronous Generators (`SG`), 2 Grid-Forming inverters (`GFM`), and 3 Grid-Following inverters (`GFL`) are immediately transformed into stratified positive (+) and negative (−) feature exemplars tracking varying distortion classes (phase jitter, sub-synchronous oscillations, and critical loop chaos). When an active validation cycle fires, high-level decision trees execute recursive structural lookups over the parts of the area:

$$\text{Area_Stable}(X) \longleftarrow \forall Y \left(\text{part_of}(Y,\, X) \longrightarrow \text{Device_Stable}(Y) \right) \tag{6}$$

By matching the localized parameter splits $(\theta_{\text{phys}}, \theta_{\text{NN}})$ across all 39 buses, the forest isolates anomalous nodes behind independent virtual LAN gateways and triggers corrective dialogue rounds locally, certifying global system-level performance while maintaining complete black-box parameter privacy.

3.9 Cross-Domain Isomorphism and Typological Analogies

We scale this architecture to support cross-domain reasoning by analogy, demonstrating that the structural form of an optimization problem applies with identical integrity across physical, social,

and spiritual lattices. When the intensional type-label stripping operator purges localized name tags, disparate realities map to identical geometric paths within the decision forest. This allows the system to unify physical SCADA grid surges, John Searle's institutional status assignments, and scriptural-spiritual salvific extractions (such as mapping Judith's mountain crevice rescue directly to the non-convex trap coordinates of the Prodigal Son). God's perspective operates as the ultimate atemporal sovereign controller (W_2) at the apex root node ($\top$), breaking the closed syntax limits of local human observation planes to enforce absolute safe performance perimeters.

4 Conclusion and Future Horizons

We have presented a robust, unified Neuro-Symbolic AI architecture that successfully scales across continuous physical motion, recursive theory of mind lattices, and discrete Searle institutional speech acts. By grounding continuous neural transformations into explainable symbolic guardrails, the framework eliminates VRAM memory black-box opacity and prevents hallucinatory logic paths.

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